

The mitotic cell cycle

A Level Biology

The structure of a chromosome

A **chromosome** 染色体 is one very long molecule of DNA wound tightly around special **proteins** 蛋白质 called **histones** 组蛋白. Winding the DNA like this lets a huge length fit inside the nucleus and keeps it tidy.

Before a cell divides, its DNA is copied (this copying is called **replication** 复制). After copying, each chromosome is made of two identical copies joined together. These two copies are the **sister chromatids** 姐妹染色单体, and they are held together at a point called the **centromere** 着丝粒. The tips of each chromosome are capped by **telomeres** 端粒, which protect the ends.

Why mitosis matters

Mitosis 有丝分裂 is a type of nuclear division that makes two **daughter cells** 子细胞 that are **genetically identical** —they carry exactly the same **genes** 基因 as the parent cell and as each other.

This matters for:

- **growth** of **multicellular** 多细胞 **organisms** 生物体 (making more cells).
- **replacement** of damaged or dead cells.
- **repair** of **tissues** 组织 by making new cells.
- **asexual reproduction** 无性生殖 (one parent makes identical offspring).

The mitotic cell cycle

The **cell cycle** 细胞周期 is the full life of a cell from one division to the next. It has three parts:

1. **interphase** 间期—the longest part. The cell grows in the **G** phase, copies its DNA in the **S** phase (replication), and grows again and prepares to divide in the **G** phase.
2. **mitosis** —the nucleus divides into two identical nuclei.
3. **cytokinesis** 胞质分裂—the rest of the cell splits, giving two separate daughter cells.

Telomeres and the ends of chromosomes

When DNA is replicated, the copying cannot reach the very end of the molecule, so a little is lost each time. Telomeres are short, repeated lengths of DNA at the ends that carry **no genes**. Because the telomeres are shortened instead, no important genes are lost during replication.

Stem cells

A **stem cell** 干细胞 is an unspecialised cell that can keep dividing by mitosis and can **differentiate** 分化 (change) into different specialised cell types. Stem cells are the source of new cells for replacing lost cells and repairing tissues.

Uncontrolled division and tumours

The cell cycle is normally tightly controlled, so cells divide only when needed. If this control is lost, a cell may divide again and again without stopping. This uncontrolled division produces a lump of cells called a **tumour** 肿瘤.

The stages of mitosis

Mitosis runs through four stages. You should be able to recognise them in photomicrographs and slides.

Stage	What happens
prophase 前期	chromosomes coil up and become visible as two sister chromatids; the nuclear envelope 核膜 breaks down; a spindle 纺锤体 of fibres forms across the cell
metaphase 中期	chromosomes line up along the middle (the equator 赤道); spindle fibres attach to each centromere
anaphase 后期	the centromeres split; the sister chromatids are pulled to opposite ends (poles) of the cell
telophase 末期	a set of chromosomes reaches each pole; a new nuclear envelope forms around each set, making two nuclei

Cytokinesis then follows. In an animal cell the cell surface membrane pinches inwards to split the cell; in a plant cell a new wall forms across the middle. The result is two genetically identical daughter cells.